QM – II Assignment – IV (Due – 20/05/2025)

1.

Prove that the traces of the γ^μ, $\boldsymbol{\alpha}$, and β are all zero.

2. Derive the matrices γ^μ and show that they satisfy the Clifford algebra.

3.

Show that

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} = I \otimes \tau_3$$

$$\gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} = \sigma^i \otimes i\tau_2$$

where I is the 2×2 identity matrix, and σ^i and τ_i are the Pauli matrices. (The $\otimes$ notation is a formal way to write our 4×4 matrices as 2×2 matrices of 2×2 matrices.)

4. Consider the spin operator, $\boldsymbol{\Sigma}$ in the Dirac theory:

$$\boldsymbol{\Sigma} \equiv \begin{bmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{bmatrix}$$

Show that $\boldsymbol{\Sigma}$ as well as the angular momentum operator, $\mathbf{L}$ does not commute with the Hamiltonian for the free Dirac Particle, but the total angular momentum, $\mathbf{J}$ does, where $\mathbf{J} = \mathbf{L} + (\Sigma/2)$.

5. Consider the 4-current:

$$j^\mu = \frac{i}{2m}\left[\Psi^* \partial^\mu \Psi - (\partial^\mu \Psi)^* \Psi\right]$$

Show that this current is conserved ($\partial_\mu j^\mu = 0$) when $\Psi(\mathbf{x},t)$ satisfies the Klein–Gordon equation.

6.

This problem is taken from Landau (1996). A spinless electron is bound by the Coulomb potential $V(r) = -Ze^2/r$ in a stationary state of total energy $E \leq m$. You can incorporate this interaction into the Klein–Gordon equation by using the covariant derivative with $V = -e\Phi$ and $\mathbf{A} = 0$.

a. Assume that the radial and angular parts of the equation separate, and that the wave function can be written as $e^{-iEt}[u_l(r)/r]Y_{lm}(\theta,\phi)$ and show that the radial equation becomes

$$\frac{d^2u}{d\rho^2} + \left[\frac{2EZ\alpha}{\gamma\rho} - \frac{1}{4} - \frac{l(l+1)-(Z\alpha)^2}{\rho^2}\right]u_l(\rho) = 0$$

where $\alpha = e^2$, $\gamma^2 = 4(m^2 - E^2)$, and $\rho = \gamma r$.

b. Assume that this equation has a solution of the usual form of a power series times the $\rho \to \infty$ and $\rho \to 0$ solutions, that is

$$u_l(\rho) = \rho^k(1 + c_1\rho + c_2\rho^2 + \cdots)e^{-\rho/2}$$

and show that

$$k = k_\pm = \frac{1}{2} \pm \sqrt{\left(l+\frac{1}{2}\right)^2 - (Z\alpha)^2}$$

and that only for k_+ is the expectation value of the kinetic energy finite and that this solution has a nonrelativistic limit which agrees with the solution found for the Schrödinger equation.

c. Determine the recurrence relation among the c_i for this to be a solution of the Klein–Gordon equation, and show that unless the power series terminates, the wave function will have an incorrect asymptotic form.

d. In the case where the series terminates, show that the energy eigenvalue for the k_+ solution is

$$E = \frac{m}{\left(1 + (Z\alpha)^2\left[n - l - \frac{1}{2} + \sqrt{(l+\frac{1}{2})^2 - (Z\alpha)^2}\right]^{-2}\right)^{1/2}}$$

where n is the principal quantum number.

e. Expand E in powers of $(Z\alpha)^2$ and show that the first-order term yields the Bohr formula. Connect the higher-order terms with relativistic corrections, and discuss the degree to which the degeneracy in l is removed.